CS & IT ENGINEERING



Algorithm

Miscellaneous
Discussion Notes

DDP -01

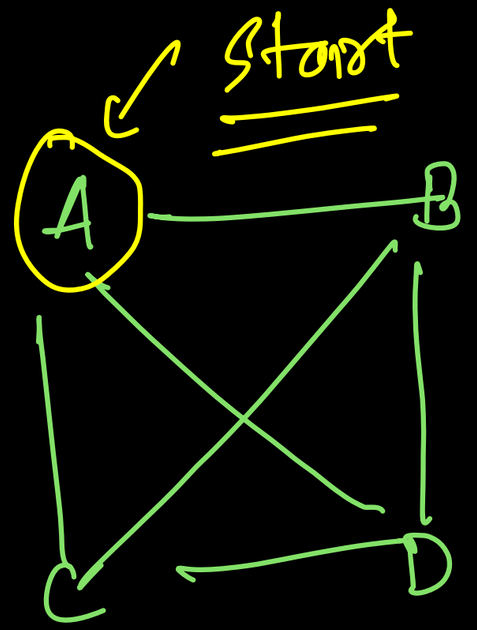


By- Aditya Sir

[NAT] **[1-Mark]**

#Q. Suppose, G is a undirected connected complete graph with 4 vertices. How many BFS traversals are possible for Graph G_____?

Ans :- 24



	B	C	D
A	3	2	1

A $3 \times 2 \times 1 \longrightarrow 6$

B $3 \times 2 \times 1 \longrightarrow 6$

C $3 \times 2 \times 1 \longrightarrow 6$

D $3 \times 2 \times 1 \longrightarrow 6$

$4 \times 6 = 24$

A B C D

A B D C

A C B D

A C D B

A D C B

A C B C

} > 6

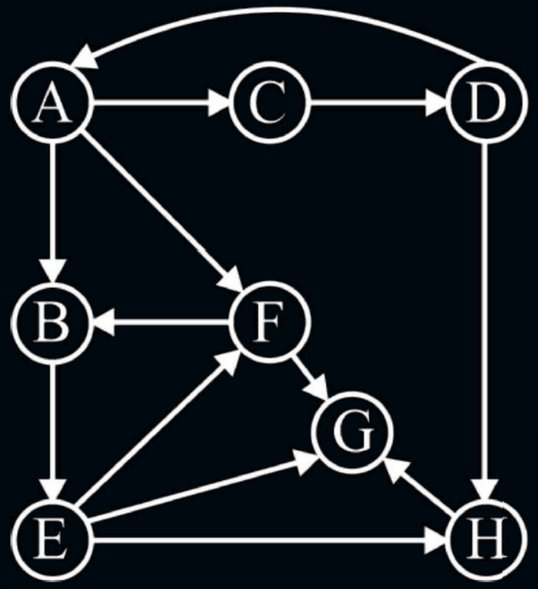
$6 \times 4 = \underline{24}$

Shortcut

$4!$

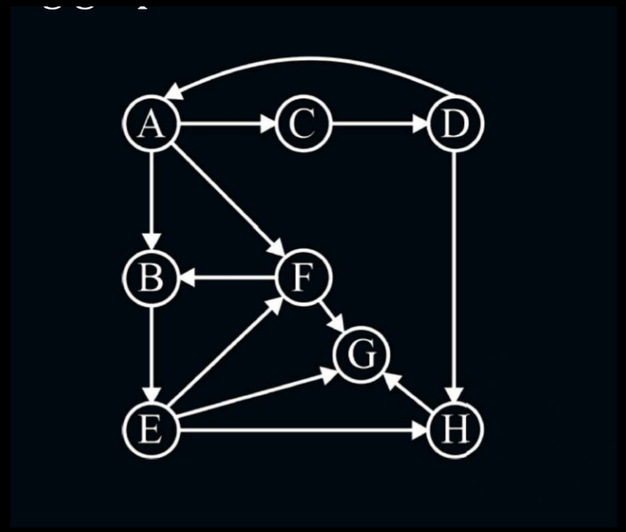
[NAT] [1-Mark]

#Q. Consider the following graph



Suppose after applying DFS traversal starting from node 'A'. How many tree edges exits_____?

Ans :- 7



DFS at A

Tree

DFT & Graph

A: 1/16
C: 12/15
D: 13/14
B: 2/11
F: 4/7
G: 5/6
E: 3/10
H: 8/9

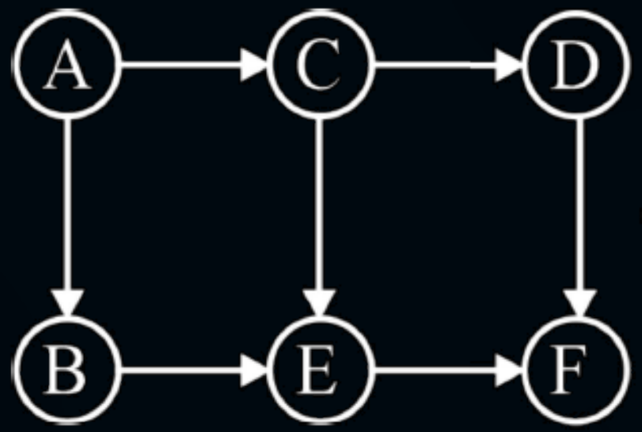
$n = 8$

$2 \times 8 = 16$

[NAT] [1-Mark]

#Q. Consider a following graph

G:



How many topological orders are possible for graph G:

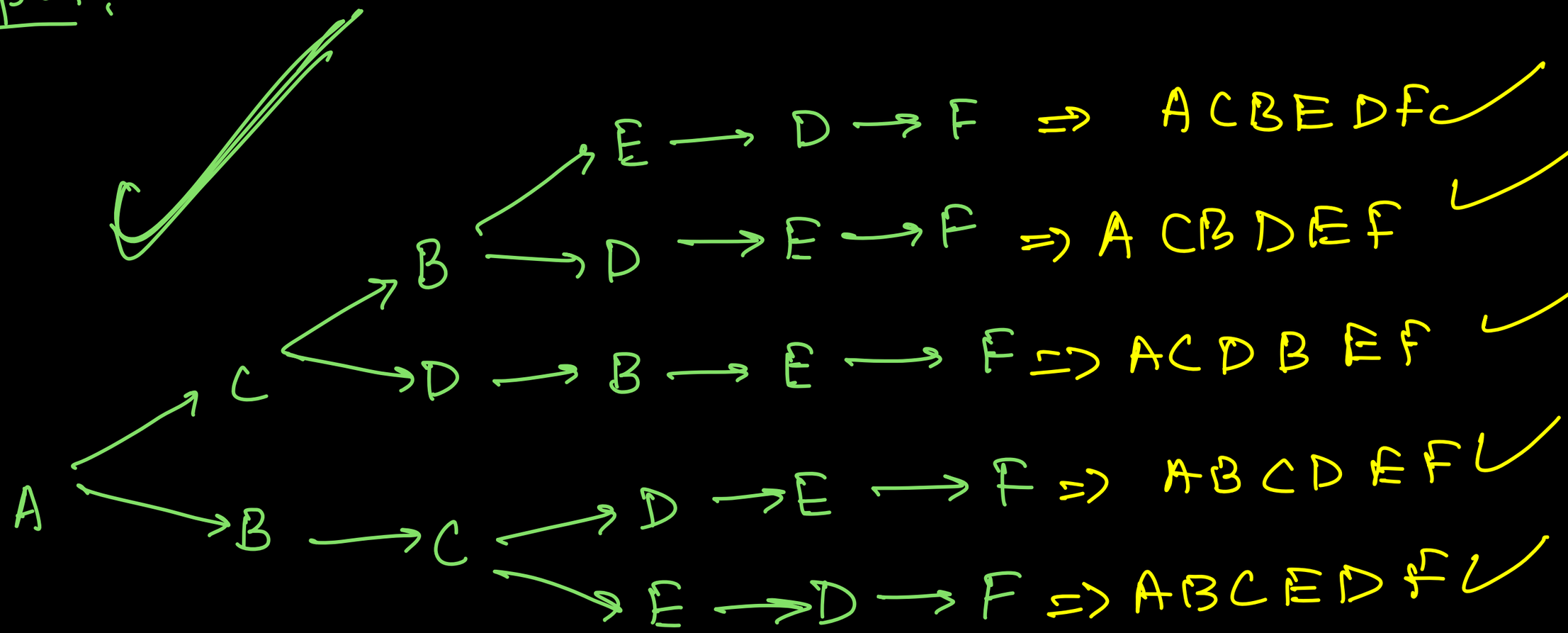
Source : A

Sink : F

Topo : A ——— F

Ans : 5

Approach :-



Shortcut:-

A _____ F

B C D E

B C E D

B D C E

[MCQ] [1-Mark]

#Q. Consider the statements

S1: Starting from vertex Vo in a graph, the time required by DFS to find a path (if exists) to some vertex V is always less than that is required by BFS.

S2: The space required by DFS is always less than that is required by BFS

Which of the following statement is true ?

wc: O(n)

A Only S1

Ans: D

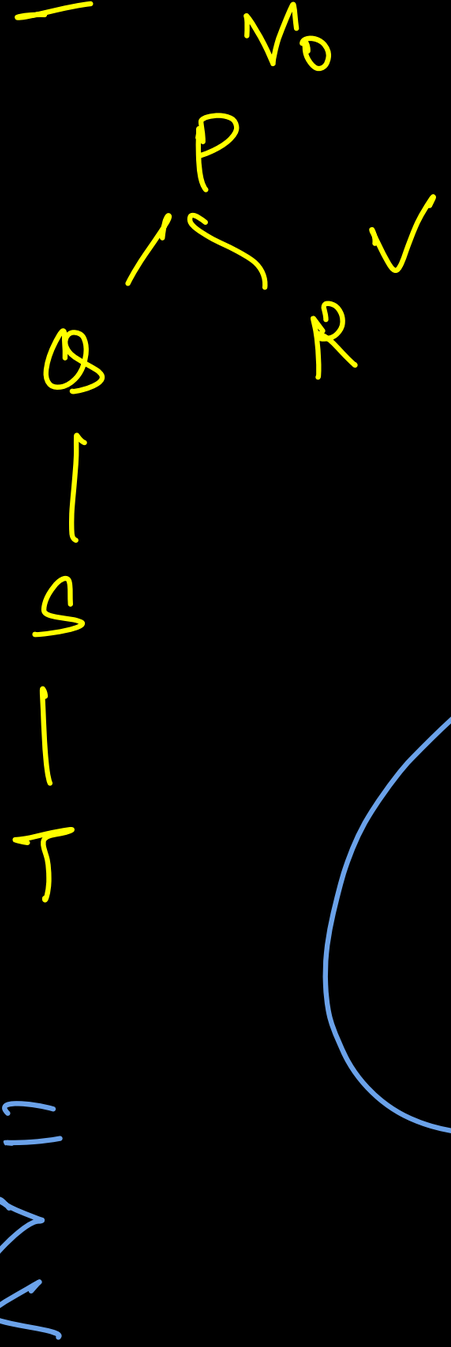
B Only S2

C Both S1 and S2

D Neither S1 Nor N2 is true

$V_0 \rightsquigarrow V$ (P ⇝ R)

Eg :-



① DFS :-

Case :- P R Q S T ⟶ 1 unit

② BFS

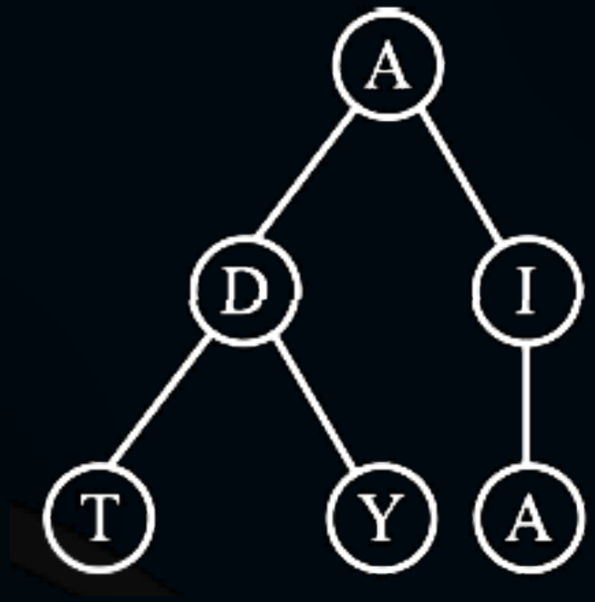
Case 2 :- P Q S T R ⟶ 4 unit

Case 1 :- 1) P R Q S T ⟶ 1 unit

Case 2 :- 1) P Q R S T ⟶ 2 unit

[MSQ] [1-Mark]

#Q. Consider the following graph



Which of the following is/are correct BFS traversal?

A, C, D

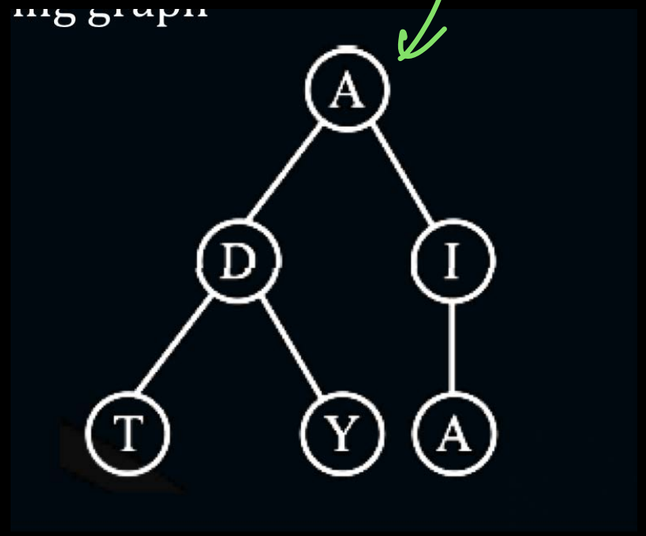
A ADITYA ✓

B AIADYT ✗

C YDATIA ✓

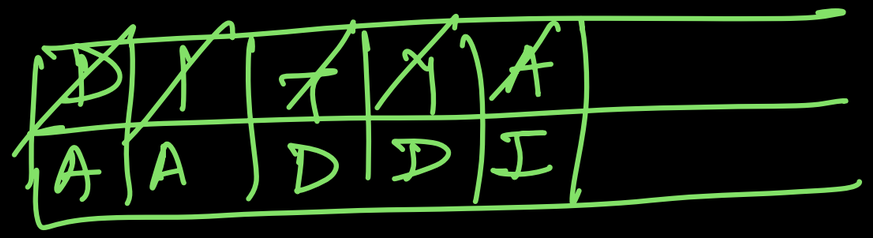
D DTYAIA ✓

ing graph

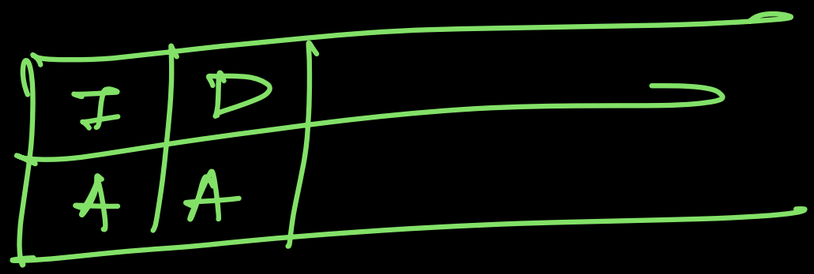


A) ADITYA

ADITYA



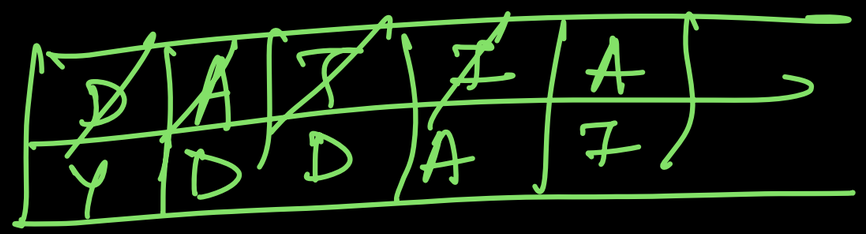
B) AIADYT



AI

C) YDATIA

YDATIA



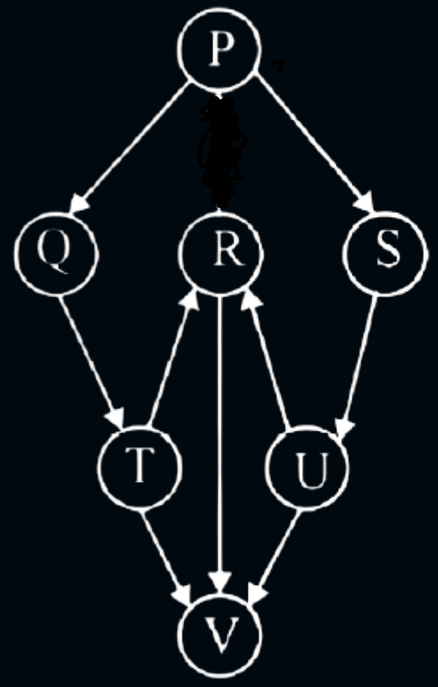
D) DTYAIA



DTYAIA

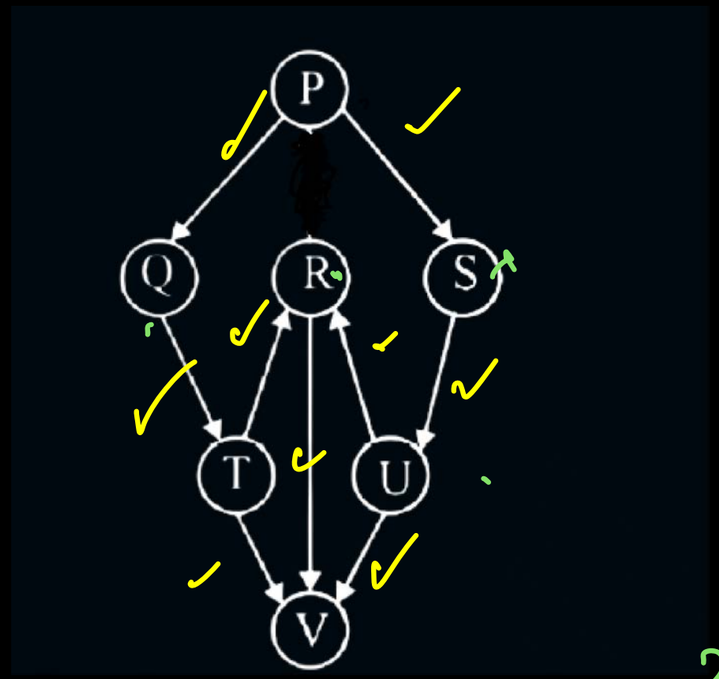
[NAT] [2-Marks]

#Q. Consider the following Graph G:



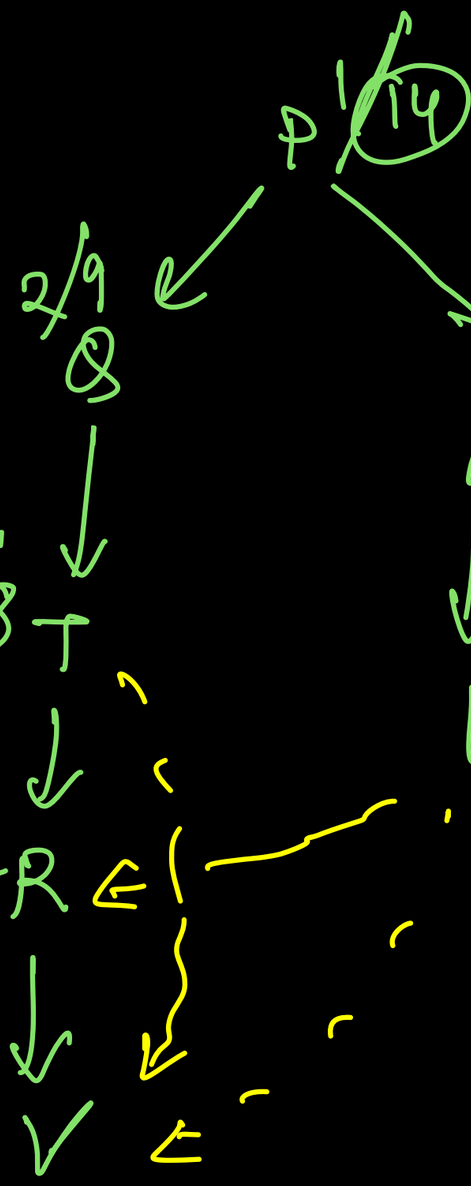
Ans :- 2

Apply DFS on G starting at vertex P and selection of adjacent vertex in DFS ~~divided~~ decided by the Lexicographical order in Graph G, Q and S are adjacent to P. First it selects Q because Q comes first in Lexicographical order. Then what is the number of cross edge when the DFS performed on G is_____.



n = 7

DFS starting at P



Types of edges:

1) Tree: (6).

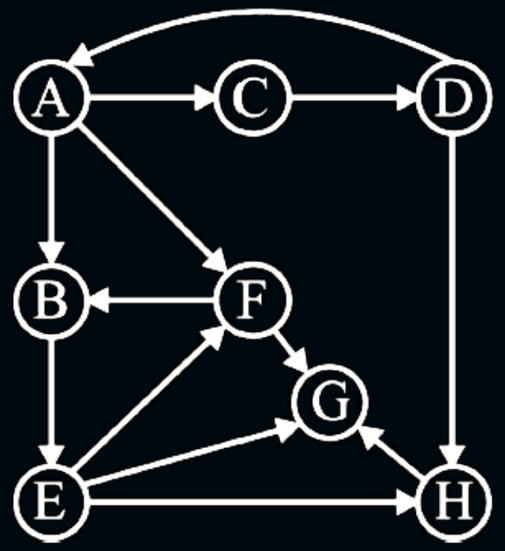
2) Forward: TV (1).

3) Back: None

4) Cross: UV, UR (2).

[NAT] [2-Marks]

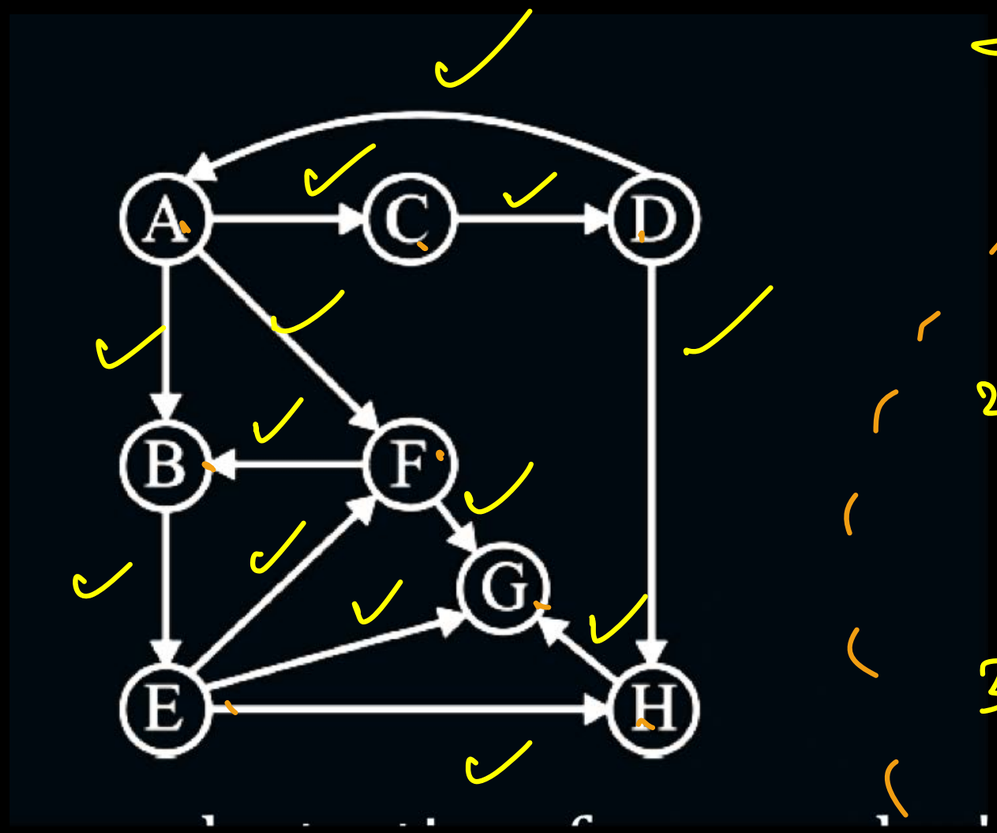
#Q. Consider the following graph



Ans: 4

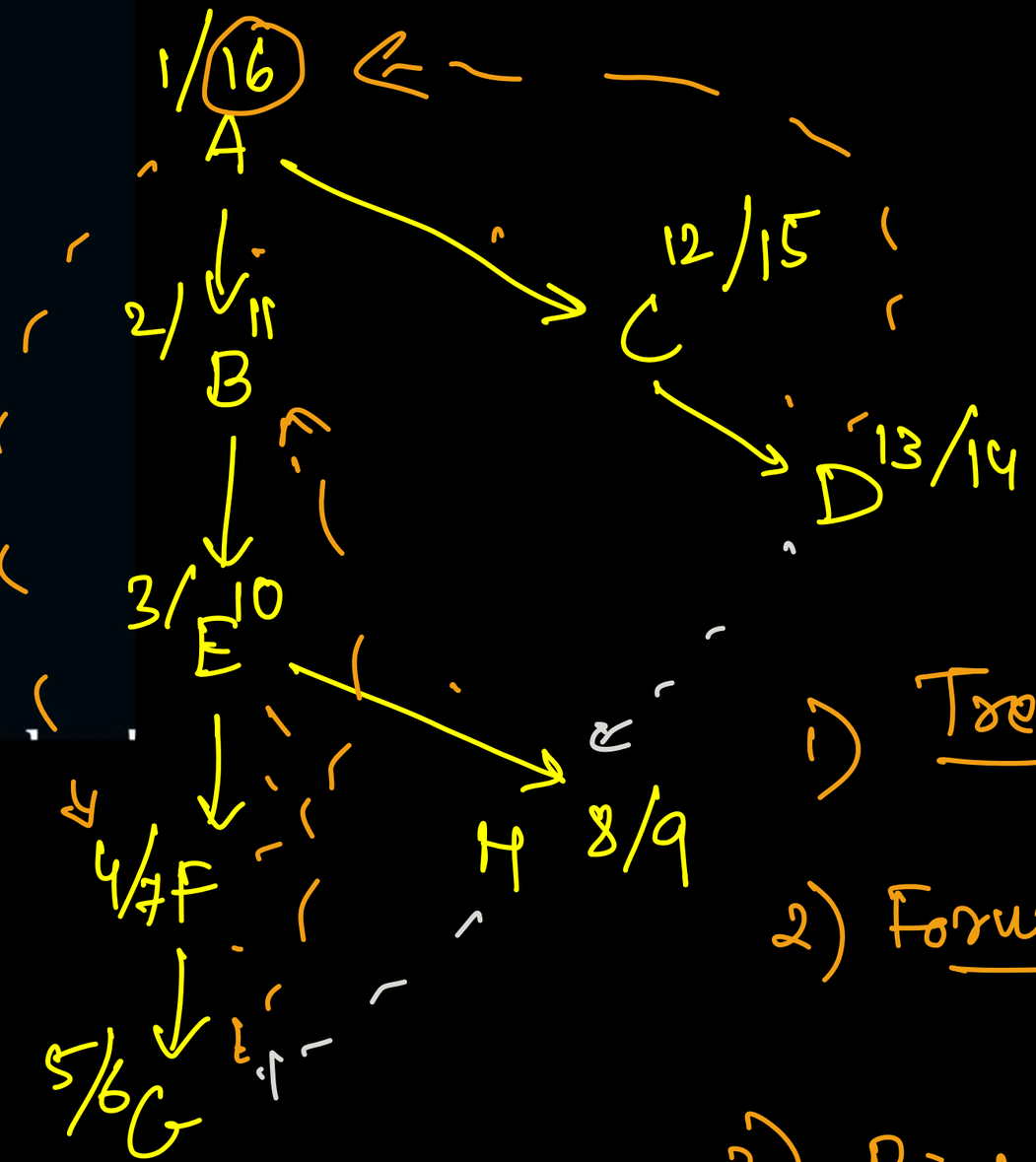
After applying DFS traversal starting from node 'A' in above graph if forward edges are x and cross edges are y then, $(x)^y$ is-

NOTE: Visit adjacent nodes in lexicographical order(B before C)



n = 8

DFS Start at A



1) Tree : 7.

2) Forward : AF, EG (2).

3) Back : DA, FB (2).

4) Cross : HG, DH → (2).

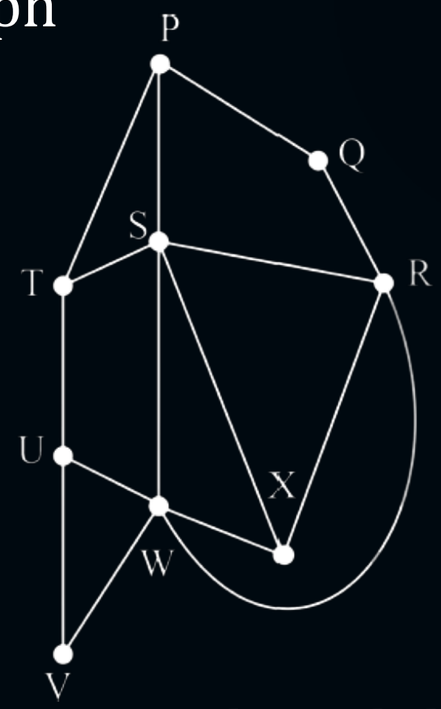
$x = 2$

$y = 2$

$x^y = 2^2 = \boxed{4}$

[MSQ] [2-Marks]

#Q. Consider the following graph



Ans :- (A, C)

Which of the following represents the valid DFS traversal?

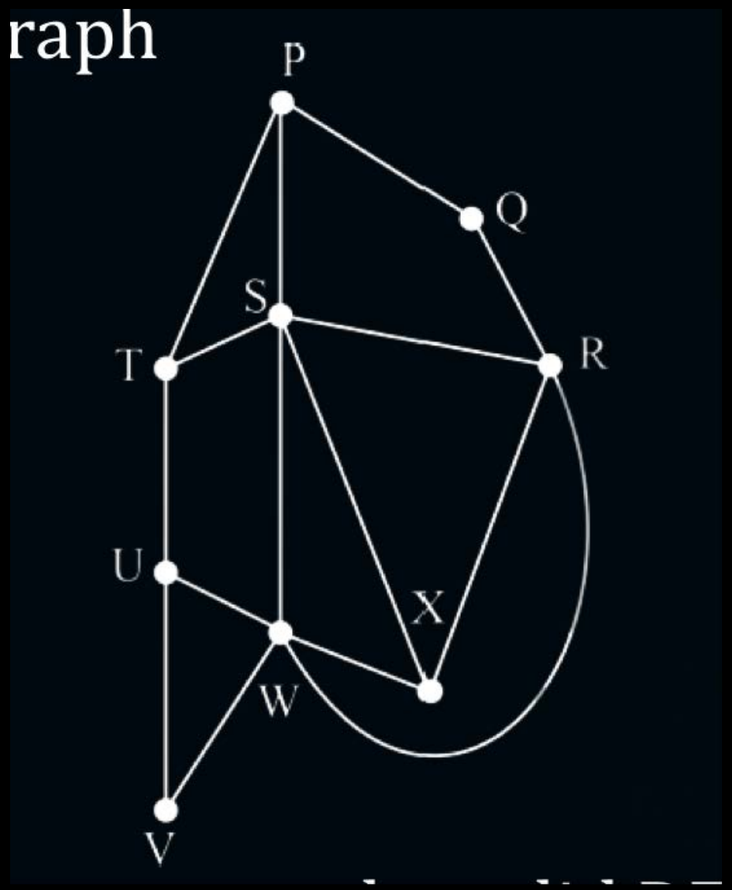
A P S W U T V X R Q

B X S P T Q R W V U

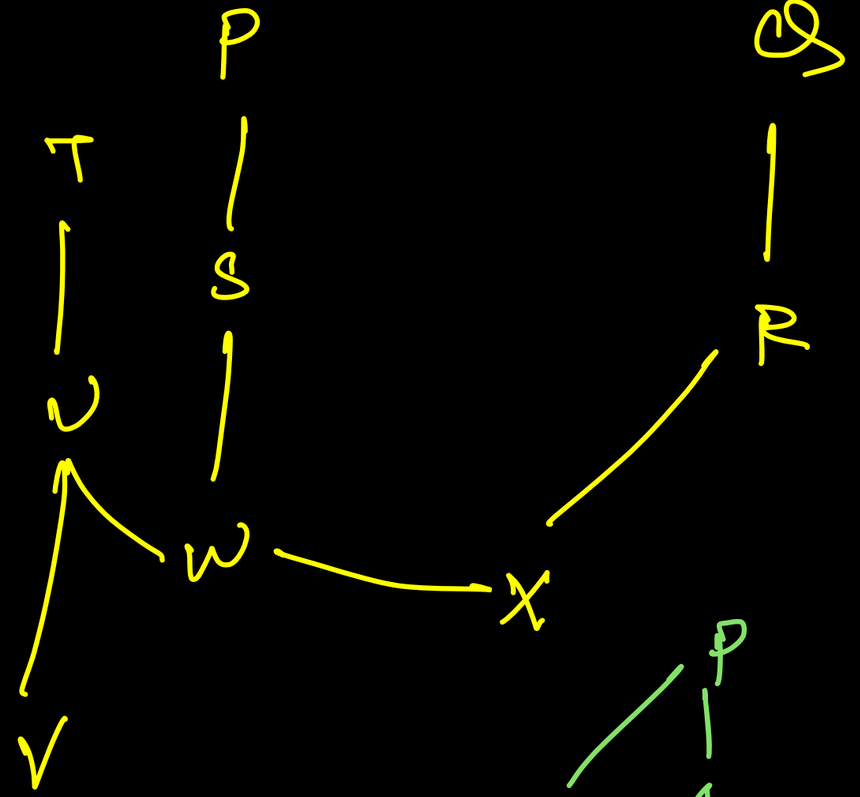
C V W X R Q P S T U

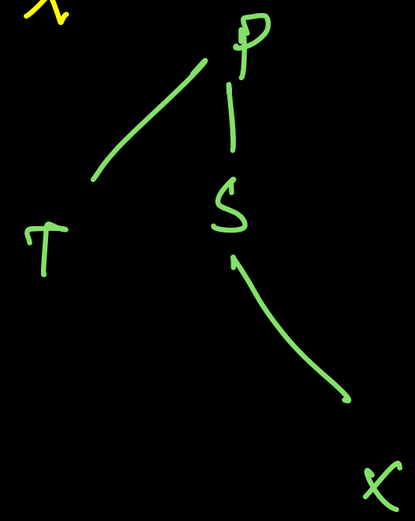
D T S P W U V X R Q

raph

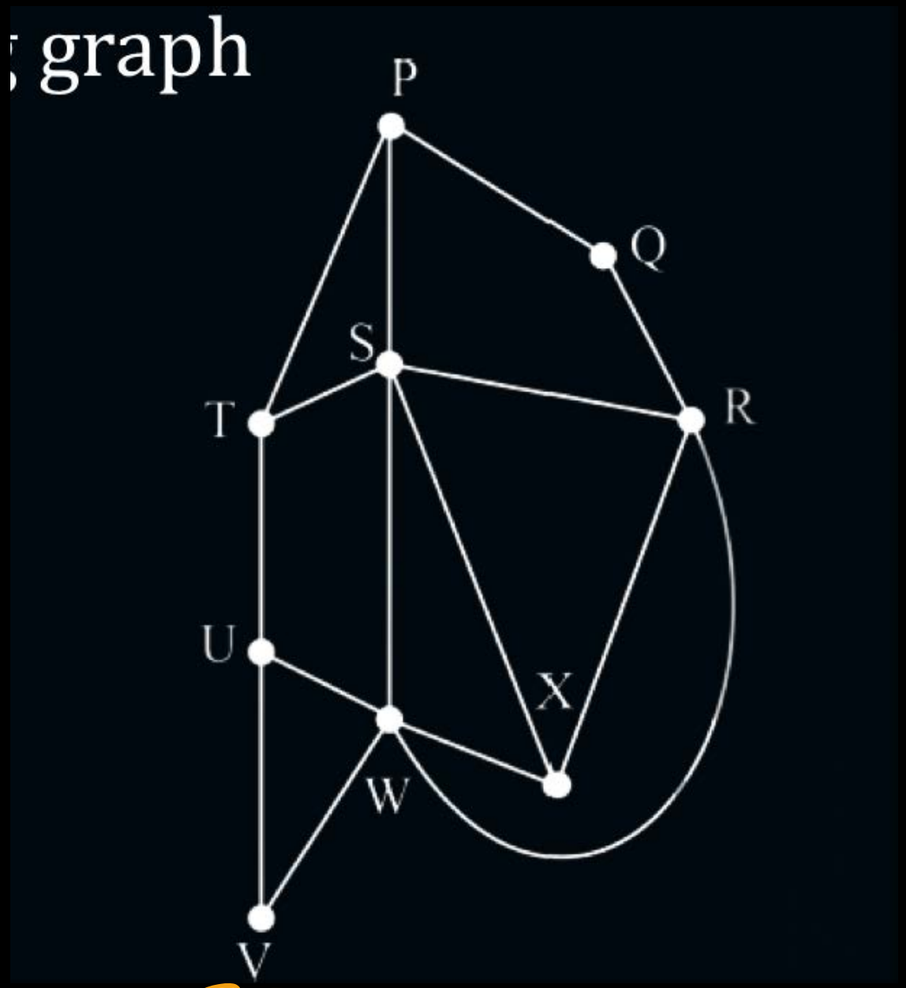


A) P S W U T V X R Q





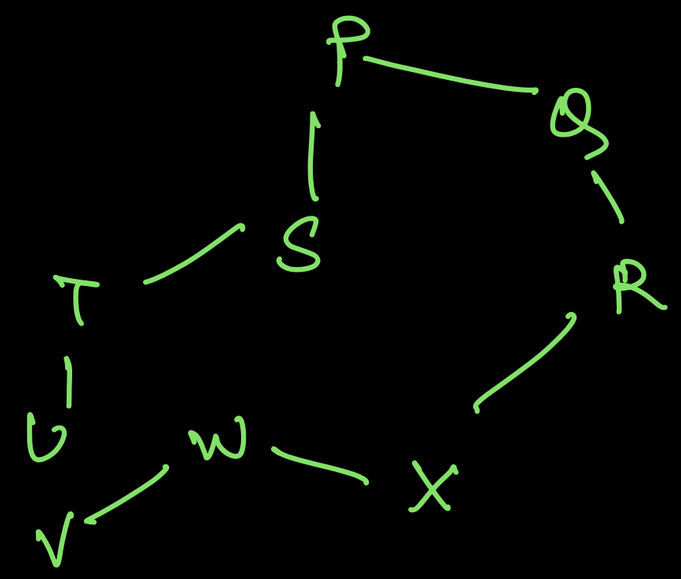
B) X S P T Q R W V U



D) T S P W U V X R Q

X

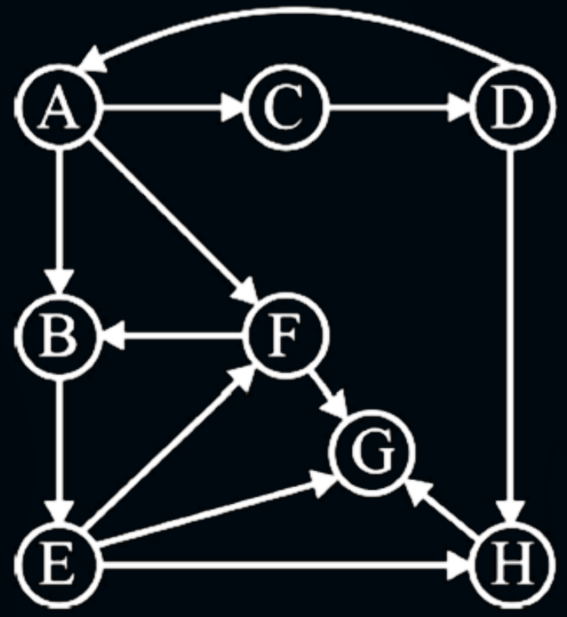
C) V W X R Q P S T U





[NAT] [2-Marks]

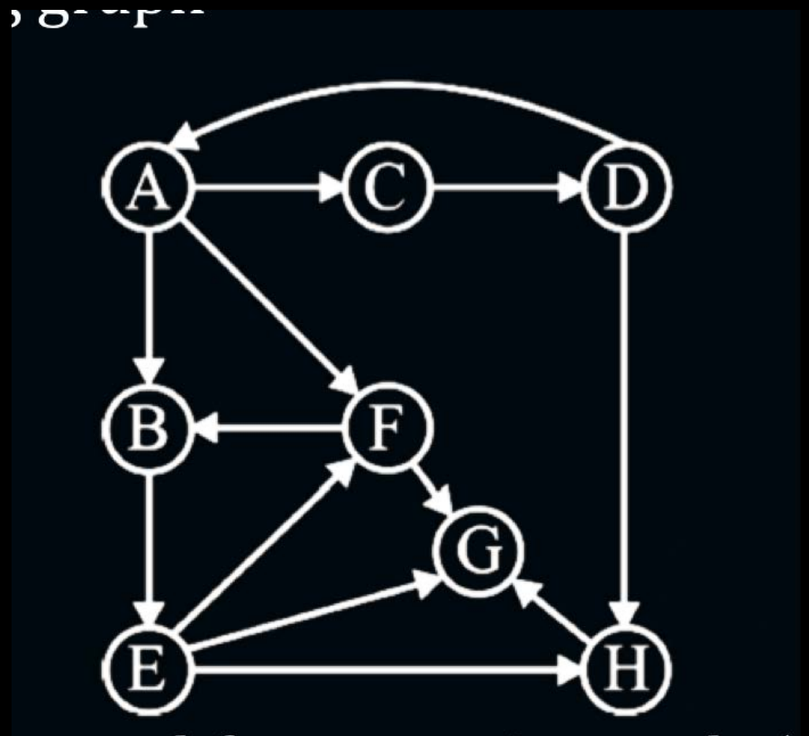
#Q. Consider the following graph

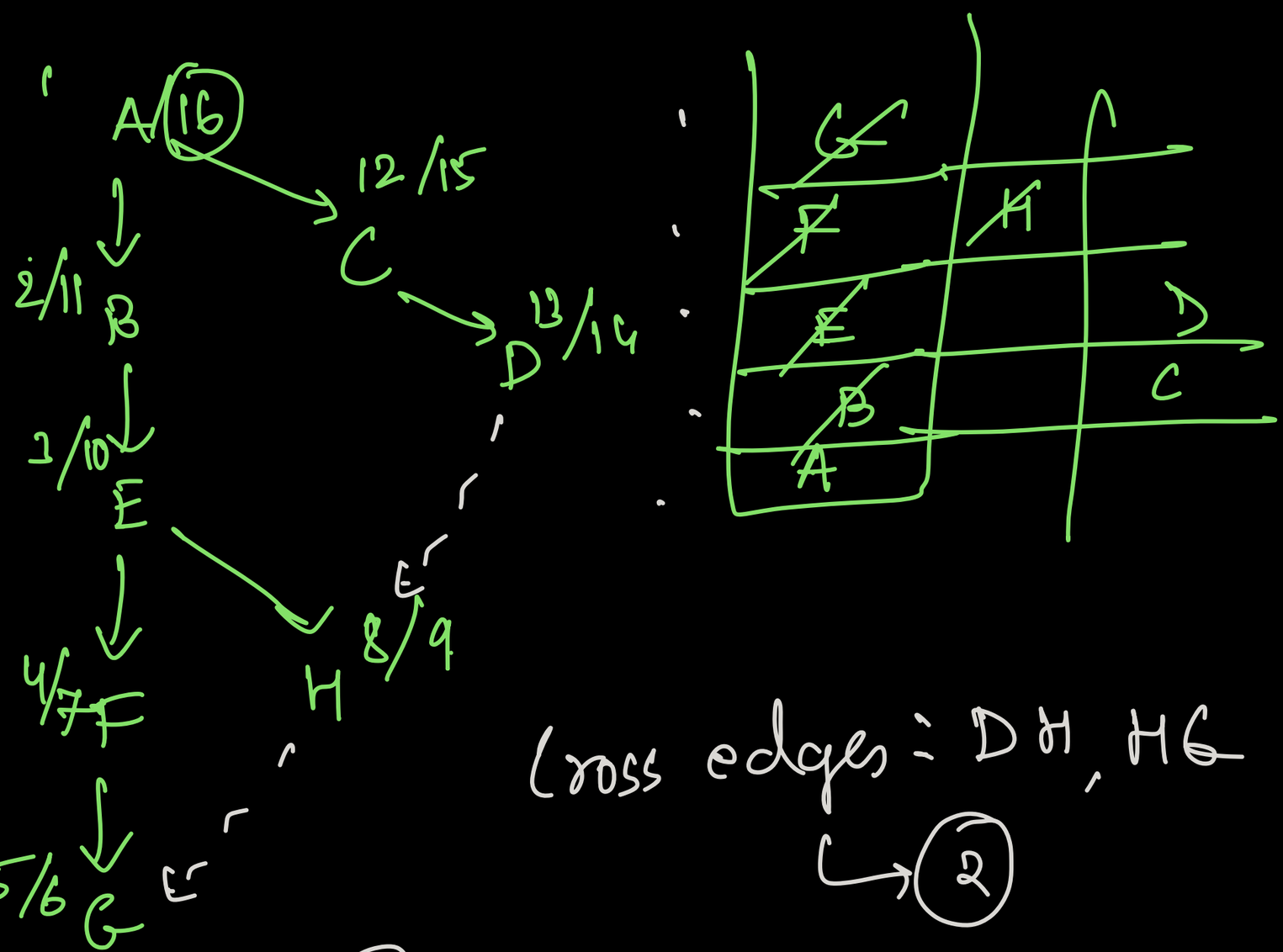


Ans: 9

After applying DFS traversal from starting node 'A' . If maximum stack size during DFS is P and number of cross edges are 'Q' then P + 2Q is___.

NOTE: Visit adjacent nodes in lexicographical order(B before C)





$n = 8$

$8 \times 2 = \underline{\underline{16}}$

Cross edges : DH, HG

(2)

$P + 2Q = 5 + 2 \times 2 = \underline{\underline{9}}$

$\Leftarrow \begin{cases} Q = 2 \\ P = 5 \end{cases}$

[MSQ] [2-Marks]

#Q. Which of the following statements is/are false?

Ans:- A, D

A. In an undirected graph, the shortest path between two nodes always lies on some minimum spanning tree

False

B. If every edge of the graph has distinct weight, then highest weight spanning tree is unique.

T

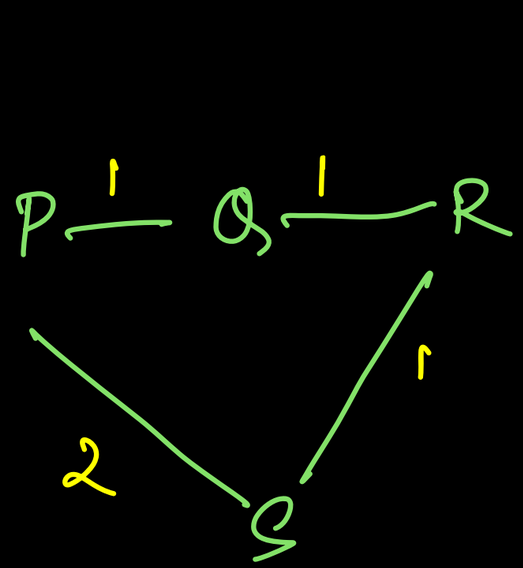
C. In Huffman coding, the item with the second lowest probability is always at the leaf that is farthest from the root

True

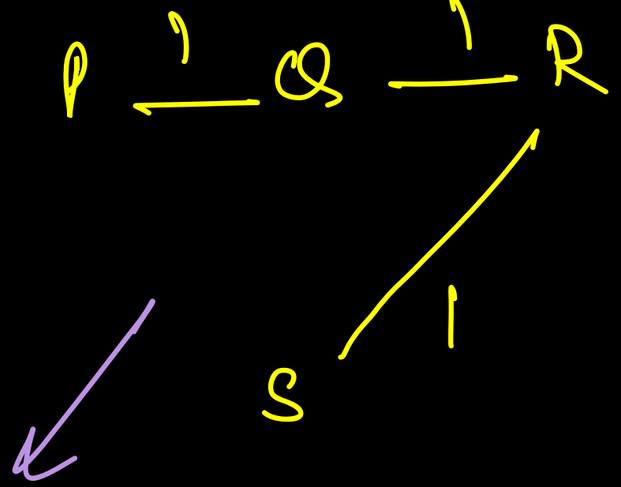
D. In Huffman coding, the item with the highest probability is always at a leaf that is the child of the root.

False

A)



MCST

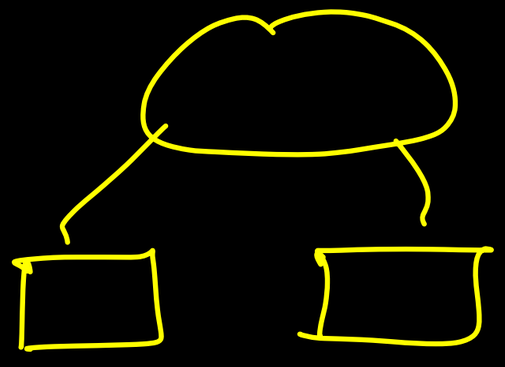


Shortest Path :

P → S : 2 ✓

P → S : P − Q − R − S

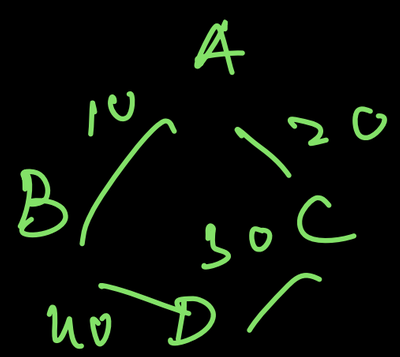
= 1 + 1 + 1 = 3



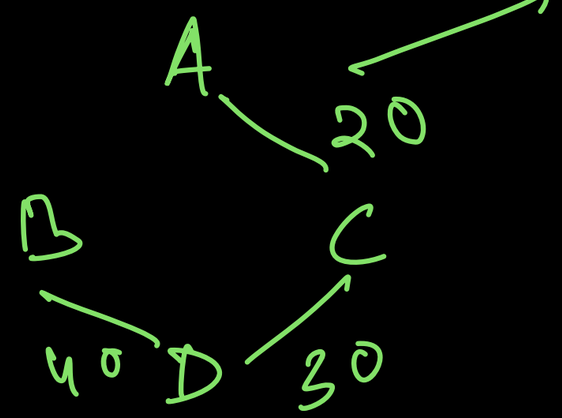
m1

m2 → lowest

(B) Kruskal



→



→ max CST

Check option D:

Qq:- 1)

P – 10 ✓

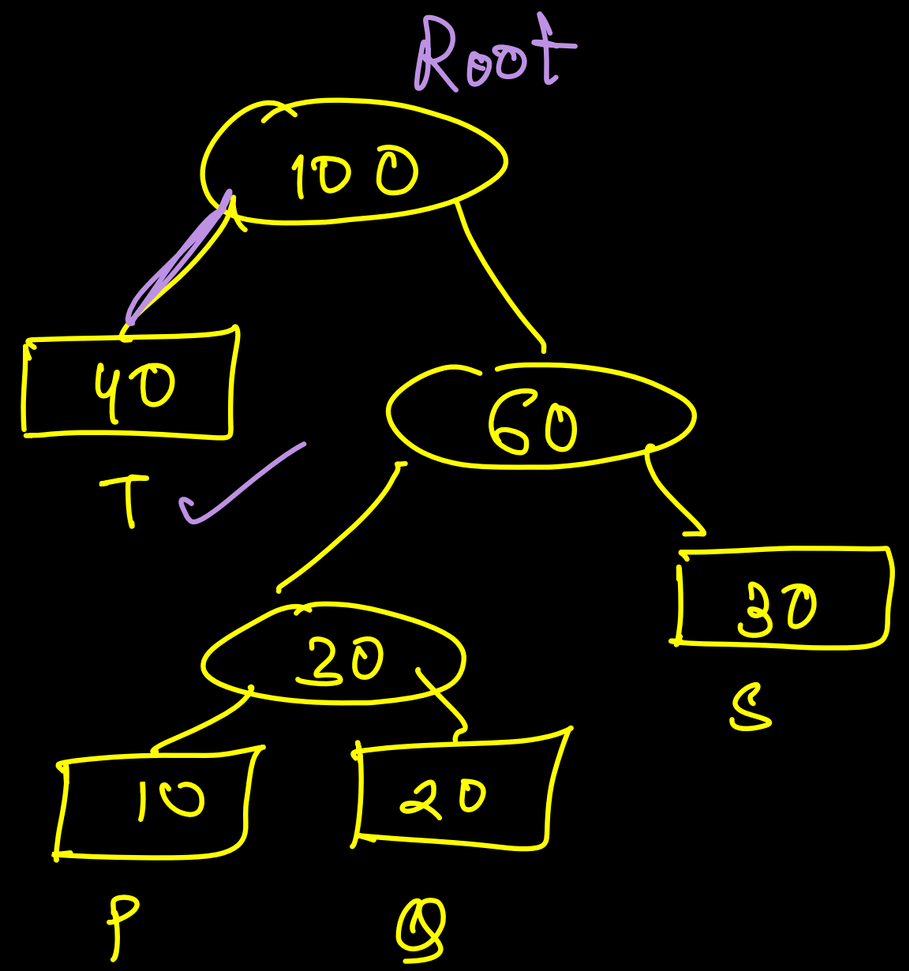
Q – 20 ✓

S – 30 ✗

T – 40 ✓

=

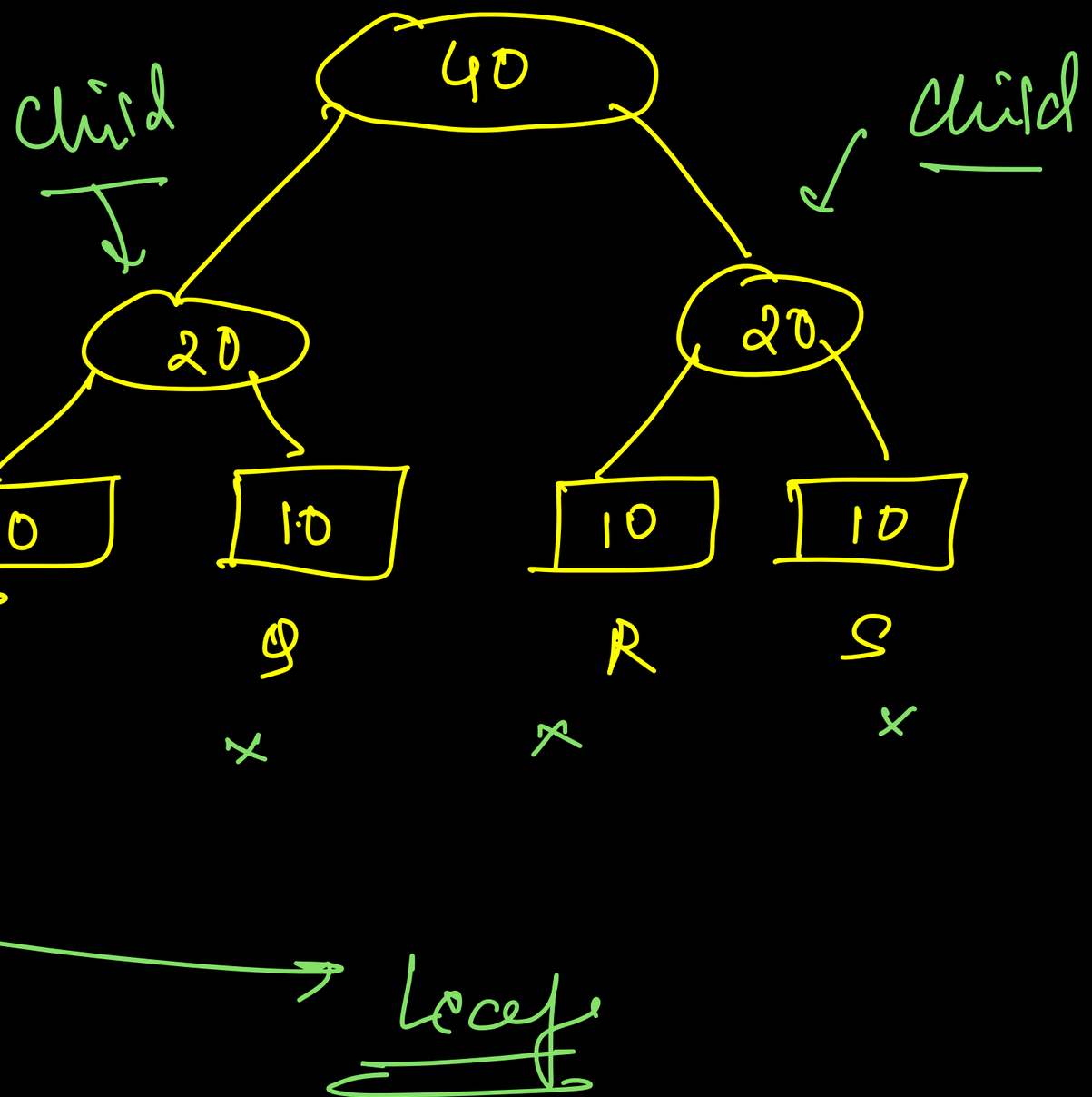
Valid



eg 2 :

P — 10
Q — 10
R — 10
S — 10

10, 10, 20, 20





THANK - YOU